EAS5450 Quiz
8:35 – 8:45AM

EAS5450
Engineering Entrepreneurship I

EAS5450 Class #5

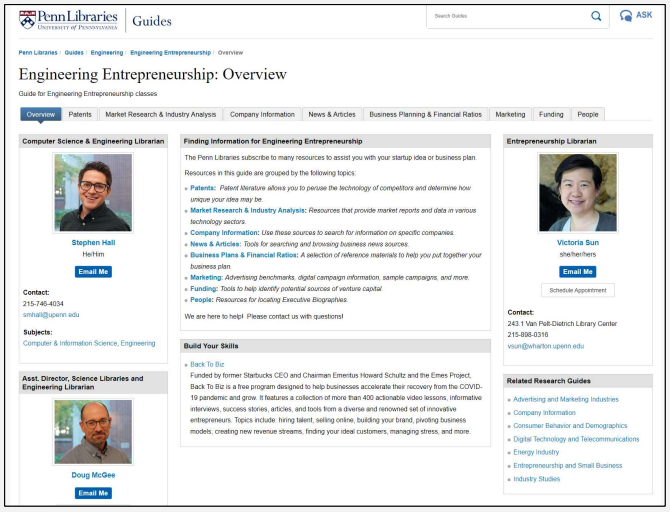
Penn Engineering

1

Library Resources Workshop

. . . . . .

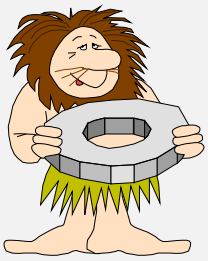
Monday, Feb 3rd
6PM
Towne 100
(EAS5450 extra credit opportunity)



2

INTELLECTUAL PROPERTY

- **Patents**
- **Copyrights**
- **Trademarks**
- **Trade Secrets**



EAS5450 Class #4

3

TRADE SECRET PROTECTION

Information used in one's business which can provide an advantage over competitors: methods, formulas, customer lists, proprietary processes, competitive intelligence.

- **Trade secrets must:**
  - Be well defined.
  - Provide economic value to the company.
  - Be subject to “reasonable measures” to keep them secret.

4

Protecting Trade Secrets

- **Non-Disclosure Agreements**
  - Cannot disclose information to third parties.
  - Not a substitute for patents
    - Does not stop 1-year clock for filing
- **Non-Compete Covenants**
  - Protects against competition from former employees
  - Nebulous legal precedent:
    - Varies from state to state
    - Courts have imposed non-competes where they did not exist, and have struck them down where they did exist.

5

WALTER KIDDE PORTABLE EQUIPMENT INC.
394 South Third Street
Mebane, NC 27302

June 8, 1999

X-IT Products, LLC
1435 Bay St., Suite 31
San Francisco, CA 94123

Dear Mr. Ive:

We have recently begun discussions concerning a potential transaction between Walter Kidde Portable Equipment Inc. ("Buyer") and X-IT Products, LLC ("Seller"). In the course of such discussions, Seller may furnish Buyer with certain information concerning the business, financial condition, operations, assets and liabilities of Seller or that may otherwise be "confidential" or "proprietary" (collectively, "Confidential Information"). Both parties wish to confirm their agreement concerning certain restrictions Buyer has agreed to accept as a condition to Seller's disclosure to Buyer of Confidential Information.

Buyer agrees that at all times, (i) it and its representatives (including, without limitation, its directors, officers, employees, agents or advisors) will use the Confidential Information only for the purposes of evaluating the potential transaction with Seller; (ii) it will provide the Confidential Information only to its employees involved in the potential transaction with Seller or outside advisors retained by Buyer in connection with such potential transaction and who agree to keep such information confidential; (iii) it will not disclose, in whole or in part, any Confidential Information (except to the extent Seller gives its prior written consent or other than as specifically provided for herein); and (iv) it will promptly return to Seller, upon ~~Buyer's~~ Seller's request, all copies of any Confidential Information provided to Buyer.

Unless and until a definitive agreement between Seller and Buyer has been executed, neither party will have any legal obligation of any kind whatsoever with respect to any such transaction by virtue of this agreement, except as expressly provided herein. Each party has the right, in its sole discretion, to reject any and all proposals made by the other party and to terminate discussions and negotiations at any time.

For purposes of this agreement, "Confidential Information" shall not, in any event, include any information that (i) is in the public domain (other than through a breach of this agreement by Buyer or any of its representatives) at the time of its disclosure to or use by Buyer, (ii) is in Buyer's possession prior to its disclosure by Seller is independently developed by Buyer (provided that the source of such Confidential Information was not bound by a confidentiality agreement with Seller or otherwise subject to any applicable restriction on the use of such information), (iii) is obtained by Buyer from a third party without any applicable restriction on use, including without limitation, a confidentiality agreement with Seller or (iv) is disclosed pursuant to the order or requirement of a court, administrative agency or other governmental body.

Background & purpose

Agreement

Limited scope

Definition(s)

6

Buyer acknowledges that neither Seller nor any of its representatives makes any representation or warranty, express or implied, as to the accuracy or completeness of the Confidential Information. Buyer agrees that neither Seller nor any of its representatives shall have any liability to Buyer, relating to or resulting from the use of the Confidential Information or any errors therein or omissions therefrom. Only those representations and warranties that are made in a final definitive agreement regarding the transaction, when, as and if expected, and subject to such limitations and restrictions as may be specified therein, will have any legal effect.

This agreement shall be binding on and inure to the benefit of all the parties and their respective successors, heirs, assigns and affiliates and shall be construed under the laws of the State of North Carolina.

By (Signature) 6/8/99

Sam A. Solomon
Chief Financial Officer

WALTER KIDDE PORTABLE EQUIPMENT INC.

Accepted and agreed to:

X-IT PRODUCTS, LLC

By Andrew Ive
(Signature)

A. Ive. 6/8/99
Print Signature Date

CEO
Title

No reps or warranties

Legally binding

No expiration term?

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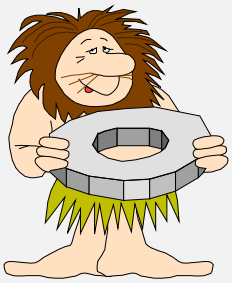
INTELLECTUAL PROPERTY

- **Do we protect it?**
- **How do we protect it?**
- **When do we protect it?**
- **Where do we protect it?**



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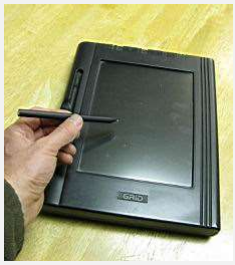
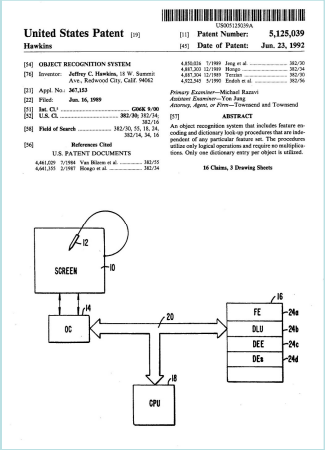


GRiD Pad



United States Patent [19]
Hawkins

[11] Patent Number: 5,125,039
[45] Date of Patent: Jun. 23, 1992

[54] OBJECT RECOGNITION SYSTEM

[76] Inventor: Jeffrey C. Hawkins, 18 W. Summit Ave., Redwood City, Calif. 94062

[21] Appl. No.: 267,153

[22] Filed: Jun. 10, 1989

[51] Int. Cl.[5] G06K 9/00

[52] U.S. Cl. 382/30; 382/34; 382/16

[58] Field of Search 382/30, 55, 18, 24, 382/34, 36, 16

[56] References Cited

U.S. PATENT DOCUMENTS

4,841,029 7/1984 Van Bilzen et al. 382/55
4,841,555 2/1987 Hongo et al. 382/54
4,850,026 7/1989 Jang et al. 382/30
4,857,202 12/1989 Hongo 382/54
4,857,304 12/1989 Terzian 382/90
4,922,545 5/1990 Endoh et al. 382/56

Primary Examiner—Michael Razavi
Assistant Examiner—Yon Jang
Attorney, Agent, or Firm—Townsend and Townsend

[57] ABSTRACT

An object recognition system that includes feature encoding and dictionary look-up procedures that are independent of any particular feature set. The procedures utilize only logical operations and require no multiplications. Only one dictionary entry per object is utilized.

16 Claims, 3 Drawing Sheets

Palm Pilot

9

10

Harvard Business School

9-D RA FT
January 19, 1996

Palm Computing, Inc. (B)

Jeff Hawkins poured over his licensing agreement with GRiD. He knew that the *device license* GRiD had sold to Microsoft had been restricted to hardware/software devices they manufactured. It was virtually worthless if the licensee was not a full system producer. He thought that GRiD's recent decision to leave hardware manufacture to others and concentrate on the software side of the business might have the same effect on his 1988 agreement with the company. If he could demonstrate that GRiD's new strategy required his consent, he would be in a much stronger negotiating position. He reexamined Sections 2.B of the licensing agreement

> Mr. Hawkins also grants to GRID a perpetual, irrevocable, exclusive, worldwide license to reproduce, distribute, and sublicense the Program *to end users in* object code form only. GRiD *may not sublicense the Program through resellers* without the prior approval of Mr. Hawkins unless such reseller is also purchasing or licensing GRiD hardware. (Italics added.)

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Harvard Business School

9-DRAFT
January 22, 1996

Palm Computing, Inc. (C)

Hawkins hammered out an agreement with Tandy Corporation that expanded the PalmPrint rights he had granted to GRiD in the 1988 license. Hawkins agreed to grant GRID a "nonexclusive, perpetual, irrevocable, worldwide license under all Intellectual Property" that carried the following new rights:

> 2.1.5 to sublicense the Program and derivative works thereof in machine-readable, source (subject to Section 12.1.6 below) and object code form through resellers if such Program or derivative work is bundled with and imbedded in other GRiD or Tandy Corporation *or their Affiliates'* applications or operating systems software. The Program and derivative works thereof may not be sublicensed unbundled to other software or hardware manufacturing companies.
>
> 2.1.6 to utilize source code of the Program and derivative works thereof pursuant to Sections 2.1.3, 2.1.4, and 2.1.5 hereunder if such source code is bundled with and imbedded in GRiD or Tandy Corporation *or their Affiliates'* applications or operating systems software; provided, however, that any such transfer or license of source code be conditioned upon the execution of a confidentiality agreement and upon the giving of notice to HAWKINS 15 days prior to any such transfer or license. Such notice shall identify the party to whom the source code will be transferred, shall specify the source code being so transferred or licenses, and shall include a copy of the confidentiality agreement to be entered into.

In return for extending the license, GRiD granted to Hawkins a royalty-free nonexclusive worldwide license:

> 6.1.1 to use the source and object code of the "C language" handwriting program, as set forth in Schedule C, and to use such source and object code to copy, translate, modify, improve, and enhance said "C" language" handwriting recognition program and to develop derivative works thereof;
>
> 6.1.2 to promote, market, and export said "C language" handwriting recognition program and derivative works thereof; and
>
> 6.1.3 to copy, have copied, have made, distribute, lease, and otherwise transfer and sublicense said "C language" handwriting recognition program and derivative works thereof in source and object code.

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6.1.4 to utilize source code of the "C language" handwriting program pursuant to Section 6.1.3 hereunder; provided, however, that any such transfer or license of source code be conditioned upon the execution of a confidentiality agreement and upon the giving of notice to GRiD 15 days prior to any such transfer of license. Such notice shall identify the party to whom the source code will be transferred, shall specify the source code being so transferred or licensed, and shall include a copy of the confidentiality agreement to be entered into.

The new agreement enabled both Hawkins and GRiD/Tandy to share information licensed to them with sublicensees, provided notice was given and confidentiality agreements were executed.

Though Tandy continued to worry that Palm might somehow use the "C language" enhancements to compete directly with its products, Hawkins' decision to allow Tandy to participate in the first round of equity financing quieted their fears sufficiently to let the deal proceed.

Hawkins saw that the Tandy investment could be very positive for Palm. Tandy's obvious interest in the Palm products made it a very likely first customer. If Tandy committed to carrying the product line, Hawkins might be able to tap into Tandy's well-established network of manufacturing partners—a very valuable entrée since Palm had not yet established a strong performance record of its own.

To ally his fears about Tandy taking over the company at some future date, Hawkins capped their total holdings at 10% and limited their participation in corporate governance to board observer status. Terms were agreed in December 1991. Palm opened for business on January 2, 1992, and the final agreement was executed on January 21, 1992.

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EPILOG

- **Cross-License Negotiated**
  - Hawkins rights to "C" Language
  - GRiD/Tandy rights to sublicense software
- **Tandy investment in Palm 1st round**
  - $300,000
  - Capped at 10% (act. 9.2%) ownership
  - Board observer only
  - Offsets potential competition from Palm
  - Palm channel to RadioShack market
  - Palm access to Tandy manufacturer network

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Product Development

Product/Market Fit:

The right product to the right customer
at the right time for the right price

16

3 questions …

- **Why will somebody buy our product?**
- **How will we deliver it?**
- **How does our company make money?**

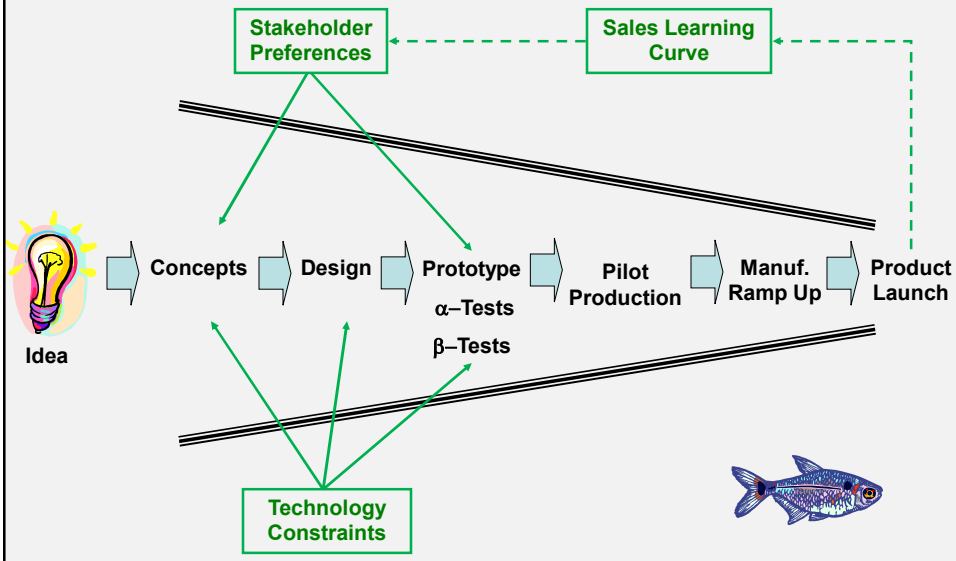
17

“Devices” ⇒ “Complete Products”

What we invent **What customers buy**

18

Product Development Funnel



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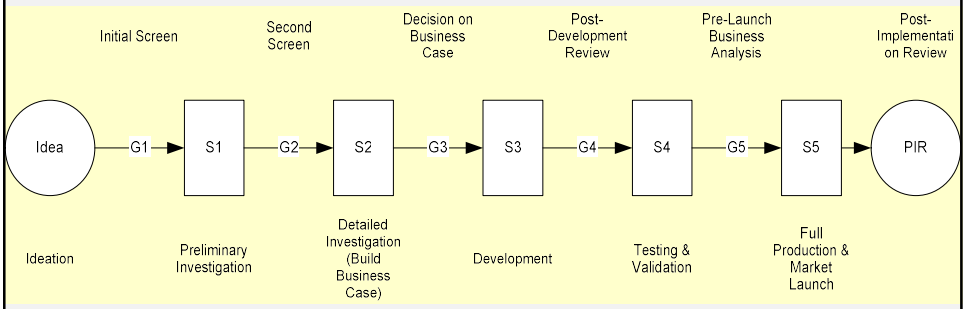
Stage-Gate Model



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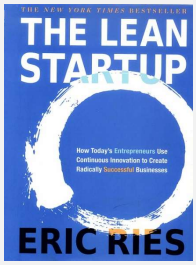
Build-Measure-Learn

Validated Learning

- Rapid scientific experimentation
  - As quickly as possible
- MVP = “minimum viable product”
- Pivot



21

Product Positioning =

Market Segmentation + Product Differentiation

1. Whose needs will be met?
2. What needs are to be met?
3. How will the product design meet these needs?

22

1. Who?

Stakeholders:

- Customers in our Target Segment(s)
- Other Stakeholders in our Value Chain:
  - Designers, manufacturers & suppliers
  - Packagers, salespeople & distributors
  - Support, maintenance & upgrade people
  - Reclamation & disposal people

23

A market segment is a group of customers sharing common desires, needs, and buying patterns.

William Davidow, *Marketing High Technology*

- Picking the right target segment is critical:
  - Identifies manageable group(s) of customers.
  - Tailors market strategy.
  - Focuses advertising and sales tactics.
  - On-going analysis supports growth.

Focus!

24

4 Criteria for Segmentation

- Psychographic homogeneity
- Identifiable
- Accessible
- Significant potential

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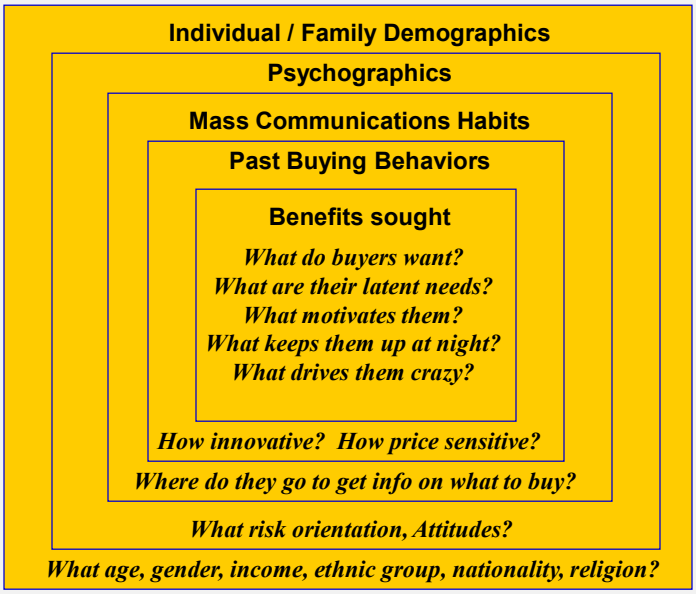
Steps for Segmenting

a) Define all potential users (the *Total Addressable Market*)
b) Understand factors of product substitution
   i. Price / Performance
   ii. Switching costs
   iii. Product differentiation
   iv. Other barriers to entry (e.g., loyalty, brand, market leader, experience, economies of scale)
   v. Other competitive threats
c) Define market segments (i.e., similar demand drivers)
d) Forecast drivers of customer demand
e) Sensitivity analysis and iterate
f) Choose the target segment

26

Example: Consumer Market Segmentation



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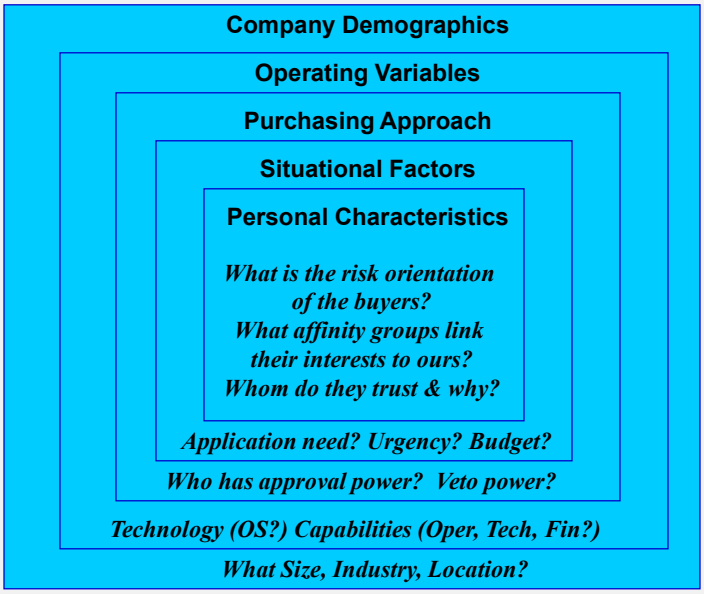
Example: Industrial Market Segmentation



28

2. What?

- Stakeholder "Preferences"
  - Relative importance of product features and benefits
  - Price-performance characteristics
  - Creating "want" for product
  - Tools available:
    - Concept testing
    - Conjoint analysis
    - Perceptual mapping

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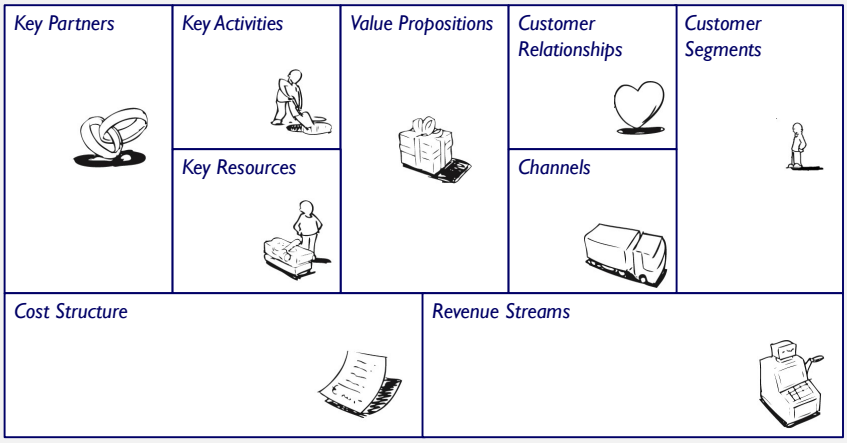
3. How?

a) Stakeholder ~~Needs~~ Wants ⇨ Product Attributes
b) Product Attributes ⇨ Concepts
c) Concepts ⇨ Design

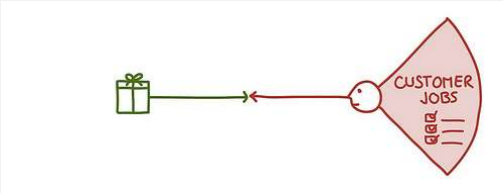
30

The Business Model Canvas



Source: Business Model Generation, Osterwalder, Pigneur et al. www.businessmodelgeneration.com

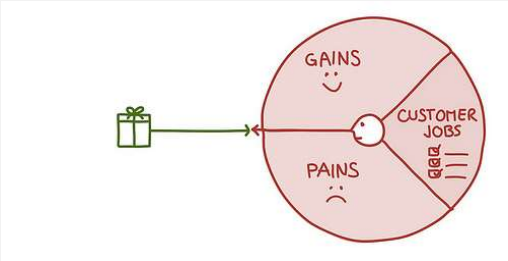
31



Functional? Social? Emotional? Basic need?

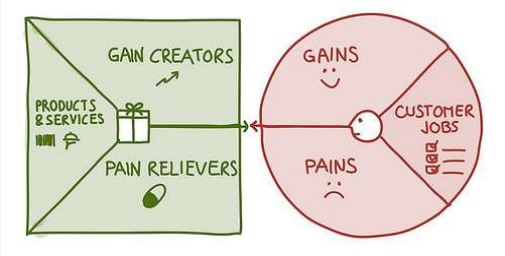
32

Benefits customer expects, desires, would be surprised by?

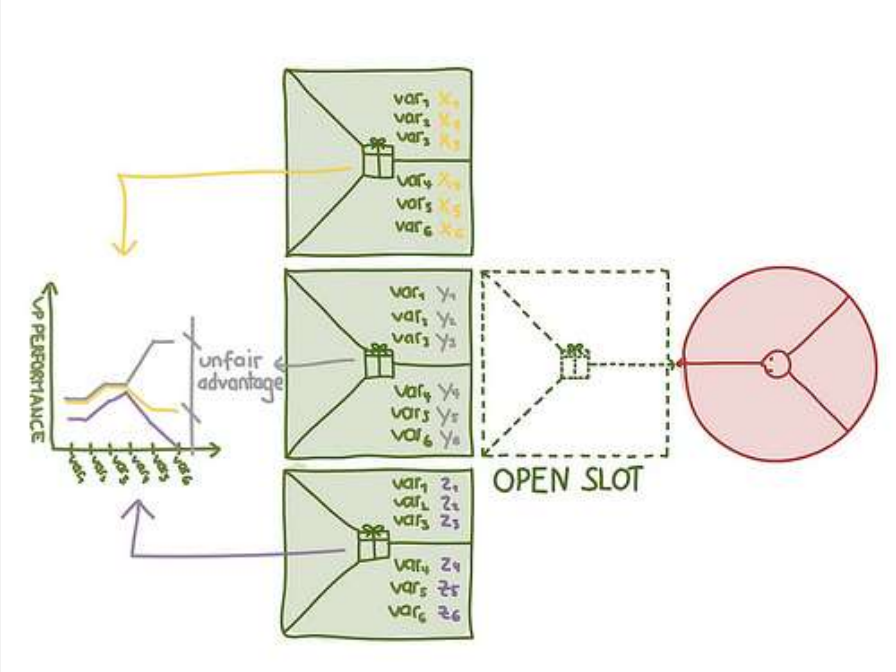


Customer pains before, during, after getting the job done?

33


GAIN CREATORS
PRODUCTS & SERVICES
PAIN RELIEVERS
GAINS
CUSTOMER JOBS
PAINS

34


VP PERFORMANCE
unfair advantage
OPEN SLOT

35

Entrepreneurial Market Research Tools

- Concept Testing
- Conjoint Analysis
- Perceptual Mapping

36

Product Positioning =

Market Segmentation + Product Differentiation

1. Whose needs will be met?
2. What needs are to be met?
3. How will the product design meet these needs?

37

Forecasting Demand

1. Define the Total Addressable Market
2. Segment the market
3. Forecast drivers of demand
4. Conduct sensitivity analyses

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Assess Market Potential

- Primary market research
- Secondary research:
  - Population-based estimates
  - Population x usage-based estimates
  - By analogy
  - By % of complementary product sales
  - Past competitor performance

39

Population x Usage Market Forecast

a)	Projected number of businesses	41,173 businesses
b)	% demand (at a given price)	× 21.2%
c)	Total potential business customers	8,728 customers
d)	No. of units per customer	× 2.9
e)	Total potential units	25,313 units
f)	% penetration	× 17.5%
g)	Estimated market demand	4,430 units in total
h)	% market share for ETPH	× 50%
i)	Estimated no. of ETPH units	2,215 units for ETPH

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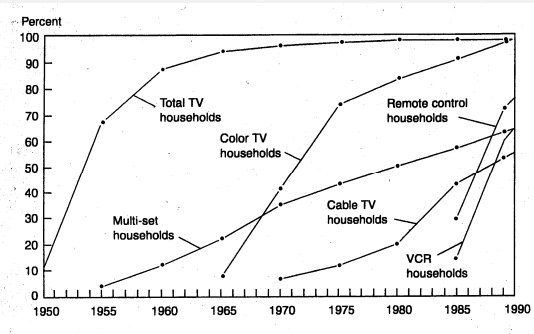
HDTV Analogous Innovations

U.S. Growth in TV and Related Innovations: 1950-1990



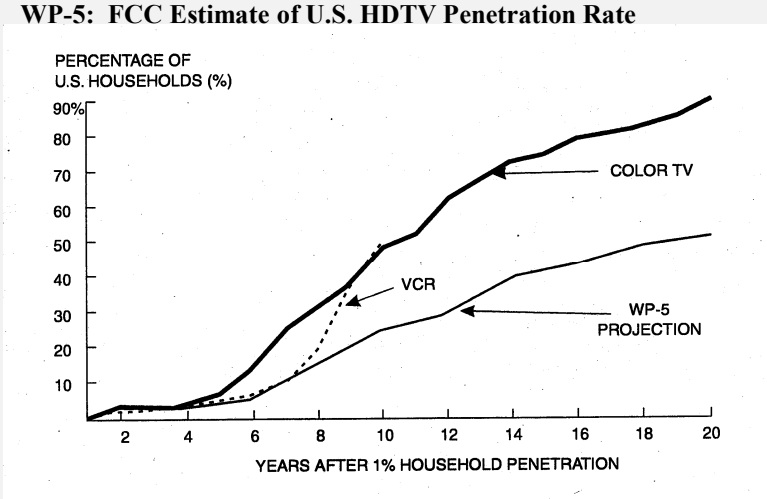
	TV households	Multi-set	Color TV	Cable	VCR	Remote control
1950	10%	-	-	-	-	-
1955	67	4%	-	-	-	-
1960	87	12	-	-	-	-
1965	94	22	7%	-	-	-
1970	96	35	41	7%	-	-
1975	97	43	74	12	-	-
1980	98	50	83	20	-	-
1985	98	57	91	43	14%	29%
1989	98	63	97	53	60	72
1990	98	65	98	56	66	77

Source: The American Enterprise, Vol. 1, No. 4, July/August 1990, p. 97, American Enterprise Institute for Public Policy Research, Washington DC.
Note: Nielsen estimated U.S. TV households = 92.1 million on Jan. 1, 1990

41

Market Forecast by Analogy

WP-5: FCC Estimate of U.S. HDTV Penetration Rate

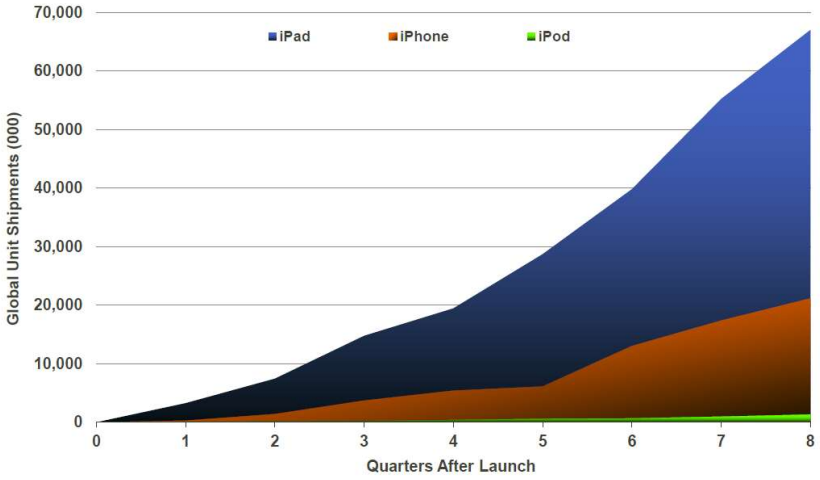


Source: Zenith

42

iPods Changed Media Industry…iPhones Ramped Even Faster…iPad Growth (3x iPhone) Leaves "Siblings" in Dust

First 8 Quarters Cumulative Unit Shipments, iPod vs. iPhone vs. iPad



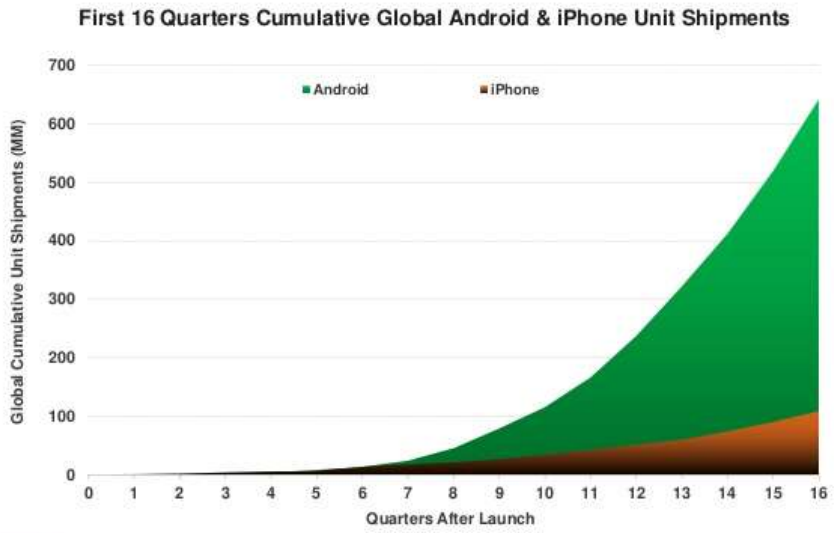
KPCB

Source: Apple, as of CQ1:12 (8 quarters post iPad launch).

9

43

Android 'Phone' Adoption Has Ramped Even Faster – Nearly 6x iPhone



KPCB

Source: Gartner, Morgan Stanley Research, as of Q3:12.

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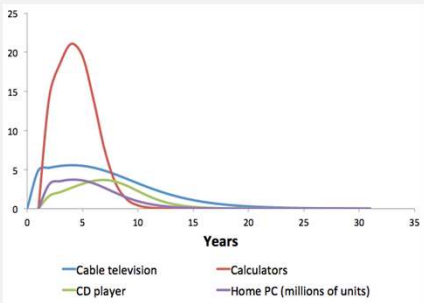
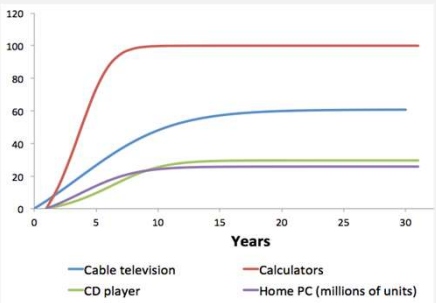
Analog Adoption Modeling





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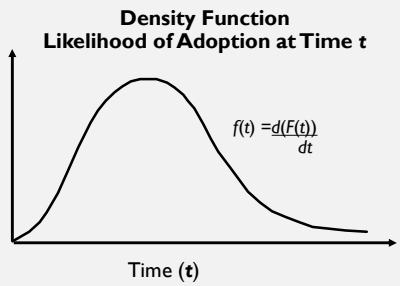
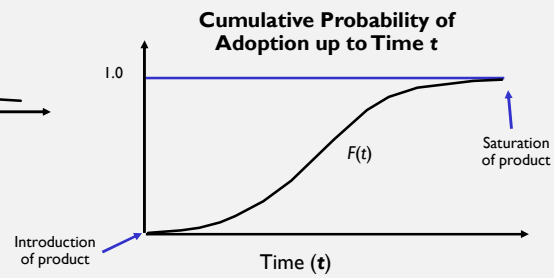
Bass Diffusion Model





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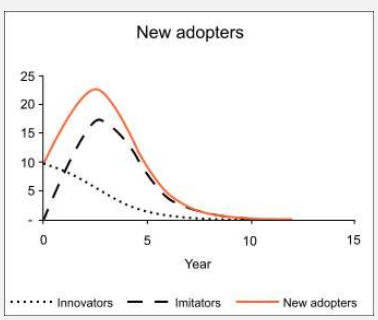

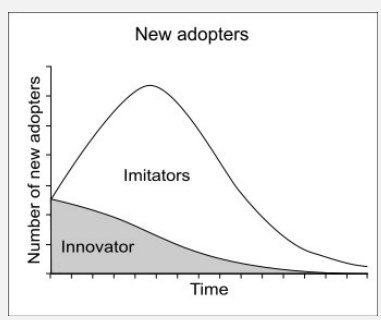

$$F(t_i) = \frac{1 - e^{-(p+q)t_i}}{1 + \frac{q}{p} e^{-(p+q)t_i}}$$

where: F(t) is the installed base fraction
t_i is the time since introduction of the ith generation product (yrs)
p is the coefficient of innovation, external influence or advertising effect
q is the coefficient of imitation, internal influence or word-of-mouth effect

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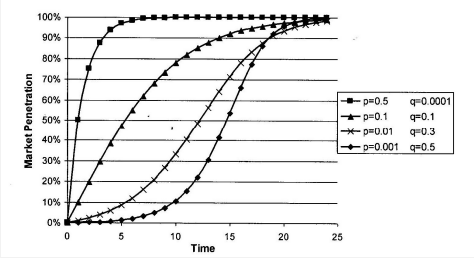
Bass Model Parameters



	Innovation	Imitation
	p (rank)	q (rank)
Calculator (1973-1979)	0.143 (1)	0.520 (7)
Home PC (1982-1988)	0.121 (2)	0.281 (14)
Cable TV Service (1981-1994)	0.100 (3)	0.060 (17)
CD Player (1986-1996)	0.055 (4)	0.378 (11)
Handheld Organizer (1997-2004)	0.043 (5)	0.565 (5)
B&W TV	0.028 (6)	0.250 (15)
Refrigerator (1926-1979)	0.025 (7)	0.126 (16)
Record Players	0.025 (7)	0.650 (4)
McDonalds	0.018 (8)	0.540 (6)
Clothes Dryers	0.017 (9)	0.360 (12)
Automobile Radio	0.016 (10)	0.410 (10)
Leaf Blower, Gas or Electric (1986-1996)	0.013 (11)	0.315 (13)
Air Conditioners	0.010 (12)	0.420 (9)
MP3 Player (1999-2004)	0.009 (13)	0.783 (3)
Cellular Telephone (1986-1996)	0.008 (14)	0.421 (8)
Color TV	0.005 (15)	0.840 (2)
Cellular Telephones	0.004 (16)	1.760 (1)

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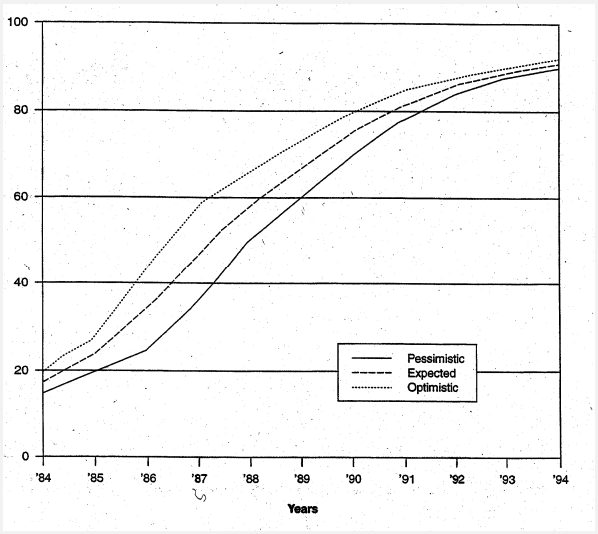
Deal with Uncertainty



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Prototyping

- Internal (“Alpha”) testing
- Stakeholder feedback (“Beta”) testing
  - *Build-Measure-Learn with MVPs*
- Communication aid

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Final Steps

- Pilot production
- Ramp up and market launch
- Feedback and improvement
- Hop to a new pond (segment)

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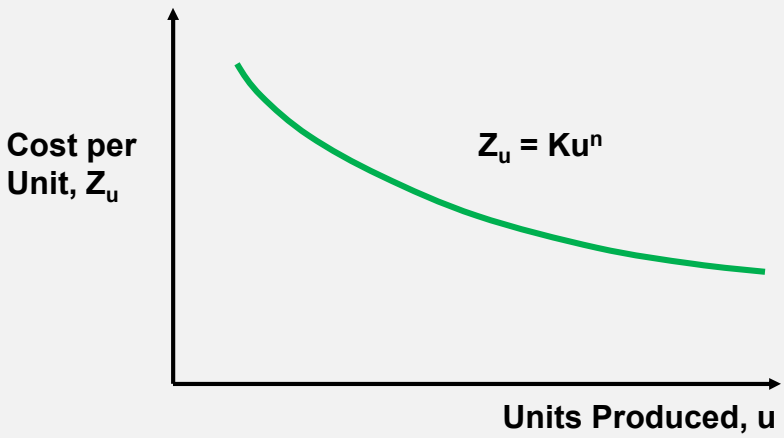
Manufacturing Learning Curve



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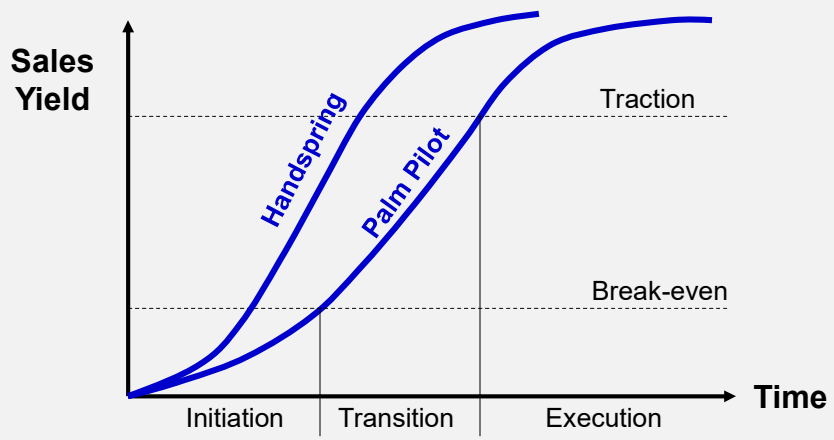
Sales Learning Curve



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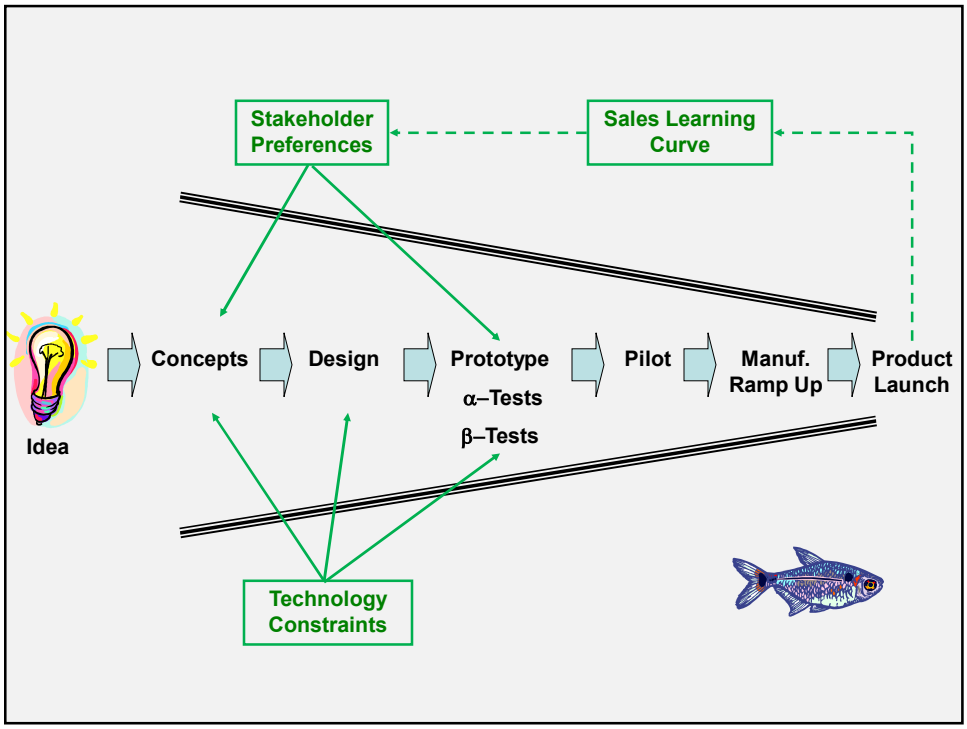

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Product-Market Fit

"Devices" ⇒ "Complete Products"

What we invent

What customers buy

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Sales & Marketing
$0-1 Million

Tyler Bosmeny

Co-Founder & CEO, Clever

https://www.youtube.com/watch?v=SHAh6WKBgiE

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The Reality…

- **It's <u>You</u>**
- Build or sell. Nothing else matters!
- Founder passion beats sales experience.

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The Almighty Funnel

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1. Prospecting

- Find the Innovators
- Top 3 methods:
  - Your network
  - Conferences
    - Go to where your customers are
    - Email attendees & set up meetings in advance
  - Cold calls/emails…

John – my name is Tyler and I'm the CEO of Clever. My company has developed new technology that reduces the time spent doing SIS integrations by 80%.

I figured this might be of interest to you given the new middle school reading software Scholastic just released.

I'd love to get your feedback even if you're not in the market for this right now. Do you have 20 minutes this week? It looks like I'm open Tuesday at 1 or 2pm ET if either may work.

Tyler

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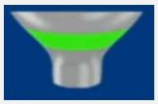

2. Conversations

- Be quiet and listen!
  - The best sales people talk <30% of the time
- Tenacious follow-up
- But, your time is precious
  - Drive to a yes (or no) ASAP

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3. Closing

- Send them an agreement
- Avoid the 3 traps:
  i. Don't quibble over minor points
     - ❑ Sign it and move on
     - ❑ Revenue is your vision
  ii. One more feature
     - ❑ Sign a conditional agreement, or
     - ❑ Wait to see if more customers want it
  iii. Free trials
     - ❑ You need a commitment, validation and revenue
     - ❑ "We don't offer free trials but can offer a 30-day cancellation period on our annual contract."

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1. Prospecting
2. Conversations
3. Closing
4. Revenue / Promised Land

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